MAT357
Foundations of Real Analysis

Class Lecture Notes

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

V Lp Spaces

16 Preliminaries

throught this section, $E \in \mathcal{E}$.

recall that $\int_E |f| = 0 \implies f = 0$ ae. this requires us to work w equiv rels of fns.

Definition 16.1

given $f, g : E \longrightarrow \overline{\mathbb{R}}$ (finite ae), we say $f \sim g$ if $f = g$ ae.

Exercise 16.2

verify that:

i) this is an equiv rel

ii) this is a well-defn'd vector space under $\alpha[f] + \beta[g] = [\alpha f + \beta g]$

one usually forgets abt equiv classes, since they rarely matter.

Definition 16.3

for $p \in [1, \infty)$, the space $L^p(E)$ is the space of (equiv classes of) fns $f : E \longrightarrow \overline{\mathbb{R}}$ st:

$$\int_E |f|^p \ < \ \infty$$

note this implies f finite ae, so $\infty - \infty$ is not a problem. if $p = \infty$, we define:

$$\|f\|_\infty \ = \ \inf \{M > 0 : m(\{x : |f(x)| > M\}) = 0\}$$

then, $L^\infty(E)$ is the set of fns w finite inf norm.

intuitively, $L^\infty(E)$ is the space of fns that are bdd ae.

Exercise 16.4

verify that:

- $m(\{x : |f(x)| > \|f\|_\infty\}) = 0$
- if $\|f\|_\infty > 0$, then $\forall \varepsilon > 0$, $m(\{x : |f(x)| > \|f\|_\infty - \varepsilon\}) > 0$

Proposition 16.5

we have the following:

i) $L^p(E)$ is a vspace

ii) $\|f\|_p = 0$ iff $f = 0$ ae

iii) $\|\lambda f\|_p = |\lambda| \|f\|_p$

Proof.

for $p < \infty$, if $f, g \in L^p$, note:

$$|f+g|^p \;\leq\; (|f| + |g|)^p \;\leq\; 2^p \left(\max(|f|^p, |g|^p)\right) \;\leq\; 2^p \left(|f|^p + |g|^p\right)$$

so $\int_E |f+g|^p \leq 2^p \left(\int_E |f|^p + \int_E |g|^p\right) < \infty$. 2 and 3 are easy. ■

Theorem 16.6: Holder's Inequality

let $p, q \in [1, \infty], p^{-1} + q^{-1} = 1$. then $|fg|_1 \leq \|f\|_p \|g\|_q$.

Proof.

we can assume $0 < \|f\|_p, \|g\|_p < \infty$, otw nothing to prove. define:

$$F(x) = \frac{|f(x)|}{\|f\|_p} \qquad G(x) = \frac{|g(x)|}{\|g\|_q}$$

so by young's inequality:

$$FG \leq \frac{F^p}{p} + \frac{G^q}{q} \implies \int_E FG \leq \frac{1}{p}\frac{1}{\|f\|_p^p}\int_E |f|^p + \frac{1}{q}\frac{1}{\|g\|_q^q}\int_E |g|^q = p^{-1} + q^{-1} = 1$$

so:

$$\int_E FG = \frac{\|fg\|_1}{\|f\|_p\|g\|_q} \leq 1$$

the case $p, q = 1, \infty$ is an exercise. ■

Theorem 16.7: Minkowski's Inequality

we have $\|f+g\|_p \leq \|f\|_p + \|g\|_p$

Exercise 16.8

prove this (its in the notes lol)

Theorem 16.9

L^p is a banach space, ie is furthermore complete.

Proof.

we'll do $p = 1$. kinda missed what he said abt the others; i think that any finite p is is just exponent juggling. forgot for inf tho. theyre both exercises still

note that for cauchy a_n:

i) if $a_{n_k} \to a$ then $a_n \to a$

ii) $\exists a_{n_k}$ st $d(a_{n_k}, a_{n_{k+1}}) \leq 2^{-k}$

pf is left as exercise. it thus suffices to show that if f_k satisfies ii), it cvgs. let:

$$F(x) \;:=\; |f_1(x)| + \sum_{k=2}^{\infty} |f_k(x) - f_{k-1}(x)|$$

then:

$$\begin{aligned}
\int_E F \;=\; \int_E \left(|f_1| + \sum_{k=2}^{\infty} |f_k(x) - f_{k-1}(x)| \right) \;&=\; \int_E \left(|f_1| + \lim_{n\to\infty} \sum_{k=2}^{n} |f_k - f_{k-1}| \right) \\
&=\; \lim_{n\to\infty} \left(\int_E |f_1| + \sum_{k=2}^{n} |f_k - f_{k-1}| \right) \\
&=\; \lim_{n\to\infty} \left(\int_E |f_1| + \sum_{k=2}^{n} \int_E |f_k - f_{k-1}| \right) \\
&=\; \lim_{n\to\infty} \left(|f_1| + \sum_{k=2}^{n} \|f_k - f_{k-1}\|_1 \right) \\
&\leq\; \|f_1\|_1 + \sum_{k=2}^{\infty} \frac{1}{2^k} \;<\; \infty
\end{aligned}$$

thus, $F \;<\; \infty$ ae, that is , $f_1(x) + \sum_{k=2}^{\infty}$ cvgs absolutely ae. therefore, the limit exists ae:

$$\lim_{n\to\infty} \left(f_1(x) + \sum_{k=2}^{n} f_k(x) - f_{k-1}(x) \right) \;=\; \lim_{n\to\infty} f_n \;=:\; f(x)$$

note that $|f(x)| \leq F(x)$ ae (apply triangle to the partial sum defn for each n). that is, $f \in L^1$. it remains to check that $\|f - f_k\|_1 \to 0$. indeed:

$$\|f - f_k\|_1 = \int_E |f - f_k|$$

note $|f - f_k| \to 0$ ae, and $|f - f_k| \leq 2F$, with $\int_E 2F < \infty$. thus, by DCT:

$$\lim_{k\to\infty} \int_E |f - f_k| = \int_E \lim_{k\to\infty} |f - f_k| = \int_E 0 = 0$$

■

consider the case $E = \mathbb{R}$. what can we say about $L^p(\mathbb{R})$? consider:

$$\|f\|_p^p = \int_{\mathbb{R}} |f|^p$$

this tells us abt the "decay rate" of f. in particular, smaller values of p are more sensitive to differences in decay rate of f. however, the other way the integral could be inf is if the fn is "spiked":

$$\int_0^1 \frac{1}{x} = \infty$$

so it also gives some way of looking at boundedness. i think. kinda missed that part

Exercise 16.10

find $f \in L^p(\mathbb{R})$ but not in $L^q(\mathbb{R})$ for all $p \neq q$.

Definition 16.11

we denote by $C_c^\infty(\mathbb{R})$ the set of smooth cptly sppted fns.

Theorem 16.12

$C_c^\infty(\mathbb{R})$ is dense in L^p for any $p \in [1, \infty)$.

we prove this for $p = 1$ (other values are similar), and by virtue of a series of lemmas.

Lemma 16.13

bdd fns of cpt sppt are dense in L^1.

Proof.

let $f \in L^1$, and define:

$$f_n(x) \ := \ f(x)\chi_{\{x:|f(x)|\leq n\}}\chi_{\{x:|x|\leq n\}}$$

then $f_n \to f$ ae and $|f_n(x) - f(x)| \leq 2|f(x)|$, so by DCT:

$$\lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \ = \ 0$$

■

Lemma 16.14

if f bdd w cpt sppt, $\exists$ seq of simples cvging to f in L^1.

Proof.

$\forall \varepsilon > 0$, $\exists$ simple ϕ st $|\phi - f| < \varepsilon$ ae. since $f \neq 0$ only on some $[-M, M]$, we can assume $\phi = 0$ outside $[-M, M]$. then:

$$\int_{\mathbb{R}} |\phi - f| \ \leq \ \int_{\mathbb{R}} \varepsilon\chi_{[-M,M]} \ = \ 2M\varepsilon$$

■

Lemma 16.15

sps $\phi \in L^1$ simple w cpt sppt. then $\exists g_n \to \phi$ cts w cppt sppt in L^1.

Proof.

suffies to consider $\phi = \chi_E$ for some $E \subseteq [-M, M]$. recall we showed $\exists O$ union of disj open int'vls st:

$$m(O \Delta E) \;=\; \int_{\mathbb{R}} |\chi_O - \chi_E| \;<\; \varepsilon$$

so suffices to approx χ_O, which we can do w trapezoids. ■

we now know cts cpt sppt fns are dense in L^1. remains to show smoothness; we modify weierstrass approx for this.

Lemma 16.16

sps f cts cpt sppt. then $\exists g_n \in C_c^\infty(\mathbb{R})$ st $g_n \to f$ in L^1.

Proof.

we know $f = 0$ outside some $[-M, M]$. let ρ be a bump fn which is 1 on $[-M, M]$ and 0 outside $[-M-1, M+1]$. then $\exists p$ poly st $|f - p| \leq \varepsilon$ on $[-M-1, M+1]$, so:

$$|f(x) - \rho(x)p(x)| \;\leq\; \varepsilon$$

and is 0 on $[-M-1, M+1]$. thus:

$$\int_{\mathbb{R}} |f(x) - \rho(x)p(x)| \;\leq\; 2\varepsilon(M+1)$$

■

last thing he does is construct ρ lol. and thats all, folks!